

# Connecting the Five-Point Relative Pose Problem to Computational Algebraic Geometry

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## 1. CONTEXT

A problem in computer vision is the reconstruction of a 3D scene. We want to find how the position and rotation of a camera changes between two images, given minimal information. Turns out that, given two static images from a previously collaborated camera, only five consistent points between images are needed for reconstruction. This five-point problem was initially proven in 1913 by Kruppa [1] to have at most eleven solutions, later improved to at most ten. This count arises as roots to a 10th degree polynomial. This paper will go through problem formation through the lens of algebraic geometry (how the 10th degree polynomial arises), computational considerations for implementation, and an example of setting up and solving the system in a computer algebra system.

## 2. PRELIMINARIES

More preliminary info can be found at [2],[3]. Define the following:

Image 1 & 2 points:  $q_1 = [x, y, 1]^\top, q_2 = [x', y', 1]^\top \in \mathbb{P}^2$

Rotation matrix:  $R$

Transform vector:  $t$

Camera matrices:

Origin (Image 1 camera matrix):

$$P_1 = [I \mid 0]$$

Shifted (Image 2 camera matrix):

$$P_2 = [R \mid t]$$

Skew Symmetric Matrix

$$[t]_\times = \begin{bmatrix} 0 & -t_3 & t_2 \\ t_3 & 0 & -t_1 \\ -t_2 & t_1 & 0 \end{bmatrix}$$

where  $[t]_\times n = t \times n$  for all  $n$

Finally, define the essential matrix, assuming calibrated camera between images:

$$E = [t]_\times R$$

The essential matrix must follow these geometric constraints:

Epipolar constraint:  $q_2^\top E q_1 = 0$

This constraint arises of geometry for points to exist in the same location in  $\mathbb{P}^3$  when projected onto the two image surfaces.

Demazure constraint:  $2EE^\top E - \text{trace}(EE^\top)E = 0$  (See [2],[3])

This constraint arises of geometry maintaining calibrated cameras.

### 3. SOLVING

The epipolar constraint is a linear polynomial in  $\mathbb{R}[E_{11}, \dots, E_{33}]$  and can be rewritten as such:

$$xx'E_{11} + yx'E_{12} + x'E_{13} + xy'E_{21} + yy'E_{22} + y'E_{23} + xE_{31} + yE_{32} + E_{33} = 0$$

For each of the 5 points. This gives a  $5 \times 9$  matrix:

$$QE = 0$$

Row reducing, we end up with:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & y'/xx' & x/xx' & y/xx' & 1/xx' \\ 0 & 1 & 0 & 0 & 0 & * & * & * & * \\ 0 & 0 & 1 & 0 & 0 & * & * & * & * \\ 0 & 0 & 0 & 1 & 0 & * & * & * & * \\ 0 & 0 & 0 & 0 & 1 & y'/yy' & x/yy' & y/yy' & 1/yy' \end{bmatrix} \begin{bmatrix} E_{11} \\ E_{12} \\ E_{13} \\ E_{21} \\ E_{22} \\ E_{23} \\ E_{31} \\ E_{32} \\ E_{33} \end{bmatrix} = 0$$

This results in a 4-dimension nullspace:

$$E = aE_1 + bE_2 + cE_3 + dE_4, \text{ with } (a, b, c, d) \in \mathbb{P}^3$$

Since  $(a, b, c, d) \in \mathbb{P}^3$ ,  $E$  is defined up to scale, so we can rescale so  $d = 1$  with  $a, b, c \in \mathbb{R}$ :

$$E = aE_1 + bE_2 + cE_3 + E_4$$

At this stage we extend  $a, b, c$  to  $\mathbb{C}$ , and return  $9 \times 1$  vector representations back to  $3 \times 3$  matrices to apply the Demazure constraint:

$$2EE^\top E - \text{trace}(EE^\top)E = 0$$

Multiplying out the  $EE^\top E$  in this constraint through SVD:

$$U\Sigma V^\top (U\Sigma V^\top)^\top U\Sigma V^\top = U\Sigma V^\top (V\Sigma U^\top)U\Sigma V^\top$$

results in a term  $U\Sigma^3 V^\top$ , so essentially we end up with 9 polynomials in  $\mathbb{C}[a, b, c]$  of maximum degree 3, which we put into an ideal:

$$I = \langle f_1, \dots, f_9 \rangle$$

We calculate the initial Groebner basis using reverse lexicographical order for computational efficiency.

Finally find the elimination ideal by changing to lexicographic order ( $a > b > c$ ). This isolates  $c$  as a 10th degree polynomial, allowing the system to be easily solved.

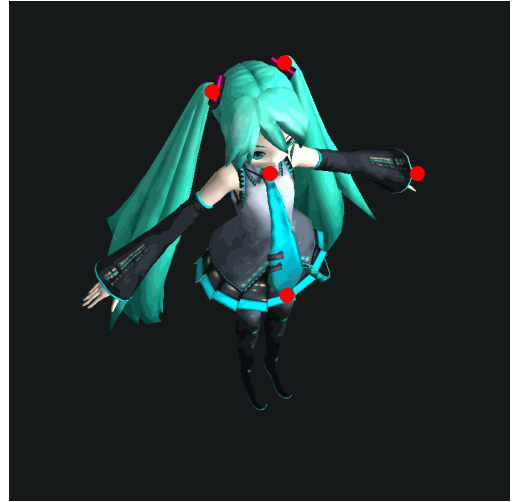
The Fundamental Theorem of Algebra guarantees this resulting 10th degree polynomial has at most 10 roots, meaning 10 total potential solutions. Another note is that after plugging in potential solutions, decomposing the essential matrix to solve for  $R$  &  $t$  brings 4 orientations per root through chirality, resulting in 40 technically possible solutions, though most are removed by geometric or physical constraints. This paper does not cover the final decomposition.

#### 4. COMPUTATIONAL CONSIDERATIONS

Buchberger's algorithm is generally slow, with worst case running around  $O(2^{2^n})$  time [4]. In application, the process to find a Groebner basis would be optimized using a  $10 \times 10$  "companion matrix" or "multiplication matrix" [2],[3]. Similar sentiment toward optimizing other methods used through the walkthrough, as this solver would ideally be run many times per second for vast amounts of points, to get an accurate sense of camera relations to space.

#### 5. EXAMPLE

For the example, we look at the following two images:



The following uses the process described in section 3, using the computer algebra system SymPy in Python to calculate.

```
# image 1
q1 = [(31, 3), (39, 6), (30, 18), (51, 44), (26, 41)]
# image 2
q2 = [(32, 14), (43, 9), (41, 27), (65, 27), (46, 44)]

# set up 5x9 matrix
Q = Matrix([
    [x1*x2, y1*x2, x2, x1*y2, y1*y2, y2, x1, y1, 1]
    for (x1, y1), (x2, y2) in zip(q1, q2)
])

# row reduce to get nullspace
E_vec = Q.nullspace()

# note we set d=1
a, b, c = symbols('a b c')
E = a*E_vec[0] + b*E_vec[1] + c*E_vec[2] + E_vec[3]
E = Matrix(E).reshape(3, 3)
```

```

# apply the Demazure constraint to get the Ideal as 9 max deg-3 polynom
I = 2*E*E.T*E - trace(E*E.T)*E
I = list(I)

# Calculate Groebner basis under grevlex order
G = groebner(I, a, b, c, order='grevlex')

# lex order for easy elimination
G = groebner(G, a, b, c, order='lex')

```

## 6. CONCLUSION AND RECOVERY

This gives us a solvable system where

$$\begin{aligned}
 a &= 0.021c^7 - 1.178c^6 + 19.777c^5 + 124.458c^4 + 562.443c^3 + 53.419c^2 + 355.918c - 2.984, \\
 b &= 0.001c^6 - 0.018c^5 - 0.112c^4 - 0.465c^3 - 0.022c^2 + 0.455c - 0.022, \\
 0 &= c^{10} - 14234.348c^9 + 2474193.001c^8 - 140746332.911c^7 + 2364672014.279c^6 + \\
 &\quad 14851727314.058c^5 + 67073452501.319c^4 + 5808873133.15c^3 + 42470573430.438c^2 - \\
 &\quad 717671811.488c + 3185875.004
 \end{aligned}$$

This has been rounded for the sake of readability. Solving numerically:

```
Poly(G[-1]).nroots()
```

We end up with 10 solutions for  $c$ :

```

[70.5879356709242
 14059.0734874288
 -2.86840360854423 - 3.54865475630046*I
 -2.86840360854423 + 3.54865475630046*I
 0.00843291162757853 - 0.00192413313323501*I
 0.00843291162757853 + 0.00192413313323501*I
 0.0204847378068047 - 0.800355996898335*I
 0.0204847378068047 + 0.800355996898335*I
 55.1826552411097 - 13.038845523214*I
 55.1826552411097 + 13.038845523214*I]

```

Of which 70.5879356709242 seems most feasible, though in other cases this could end up with more feasible solutions in  $\mathbb{R}$ .

Solution(s) would be used to get  $a, b$  then plugged back into  $E = aE_1 + bE_2 + cE_3 + E_4$  to get  $E$ . Finally,  $E$  would be decomposed into  $R$  and  $t$ .

## 7. REFERENCES

- [1] Gallego, G., Mueggler, E., & Sturm, P. (2017). Translation of “Zur Ermittlung eines Objektes aus zwei Perspektiven mit innerer Orientierung” by Erwin Kruppa (1913). ArXiv.org.  
<https://arxiv.org/abs/1801.01454>
- [2] Nister, D. (2004). An efficient solution to the five-point relative pose problem. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 26(6), 756–770. <https://doi.org/10.1109/tpami.2004.17>
- [3] Zuzana Kukelova, Bujnak, M., & Pajdla, T. (2008). Polynomial Eigenvalue Solutions to the 5-pt and 6-pt Relative Pose Problems. CiteSeer X (the Pennsylvania State University).  
<https://doi.org/10.5244/c.22.56>
- [4] Dubé, T. W. (1990). The Structure of Polynomial Ideals and Gröbner Bases. *SIAM Journal on Computing*, 19(4), 750–773. <https://doi.org/10.1137/0219053>